Evaluating Computational Prediction of HLA Class I - Peptide Binding Affinity

NetMHCpan-4.1 BA v/s experimental IEDB binding data for HLA-A*02:01 9-mer peptides

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Objectives

To understand, explore and evaluate:

- The biological basis and the algorithm behind MHC class I peptide binding prediction.
- How well computational predictions agree with experimental binding measurements.
- The association between predicted HLA class I binding strength and experimentally observed CD8+ T-cell responses.

Background

- MHC Class I (HLA Class I in humans) molecules present short peptides (mostly **9-mers**) to cytotoxic CD8+ T-cells.
- The peptide sits in a groove, usually anchored by **2nd** and **9th** residue, and the identity of the amino acids at these 2 positions strongly influences **binding strength.**
- Different HLA alleles → Different pockets: **Peptide preference of Alleles**

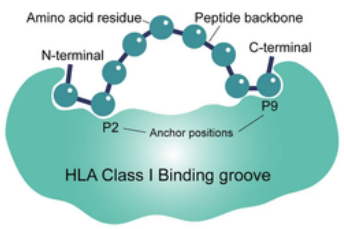
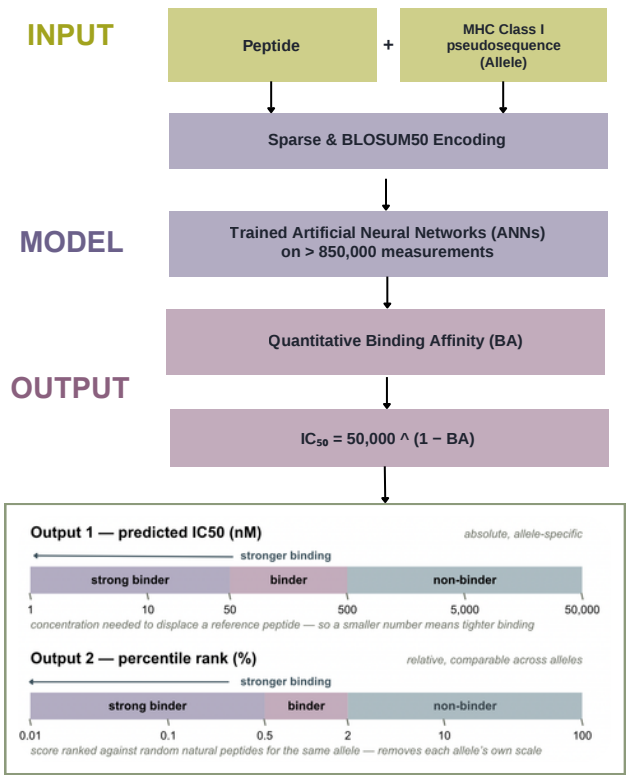


Fig 2: 9-mer peptide in the MHC class I groove, with anchors at P2 and P9 https://www.researchgate.net/

Prediction Algorithm: NetMHCpan-4.1 BA

NetMHCpan-4.1 **predicts binding of peptides to HLA molecules** of known sequence using artificial neural networks (ANNs). It is trained on **>850,000 BA and EL measurements**, with quantitative binding data from **Immune Epitope Database (IEDB).**

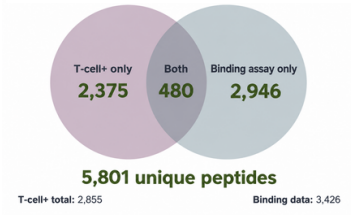


The model is **"pan-specific"** since alleles are represented by the amino acid sequences that line the peptide-binding groove

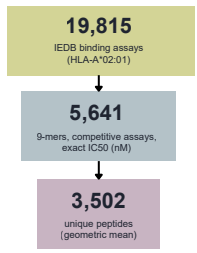
Data and methods

Source: IEDB | Human HLA-A*02:01 | Linear 9-mer peptides

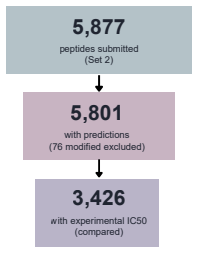
Dataset Composition



Binding Data

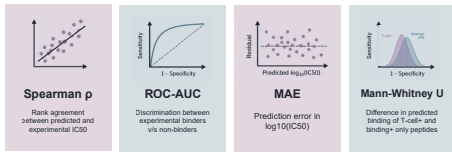


Prediction



Model selection: NetMHCpan-4.1 BA was used because it predicts quantitative binding affinity (IC50), enabling comparison with experimental measurements.

Statistical Evaluation



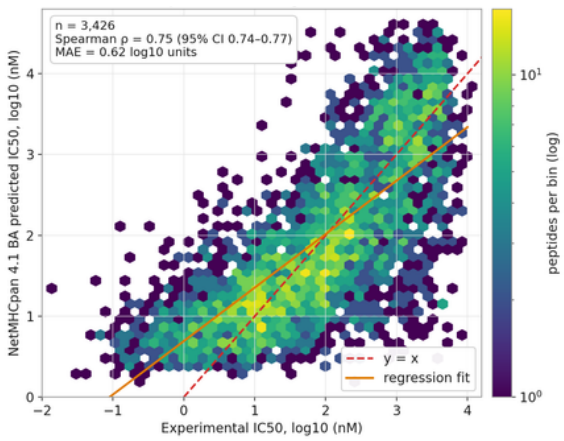
Binding classification thresholds

IC50 < 500 nM	IC50 < 50 nM
Binder	Strong binder

Lower IC50 indicates stronger binding affinity.

Results

Predicted vs. experimental IC50 for HLA-A*02:01 9-mers (n = 3,426)

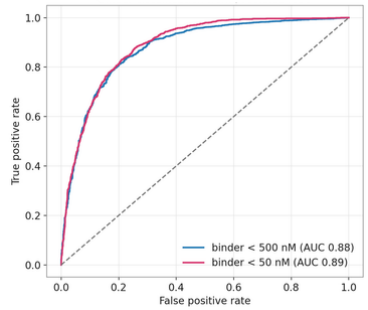


ρ = 0.75
Spearman rank correlation

MAE = 0.62
(log_{10} units)
≈ 4-fold typical error

Strong rank agreement. The fitted slope (0.66) is below 1, so predictions are compressed toward the middle of the range.

ROC curves for binder classification at 500 nM and 50 nM thresholds

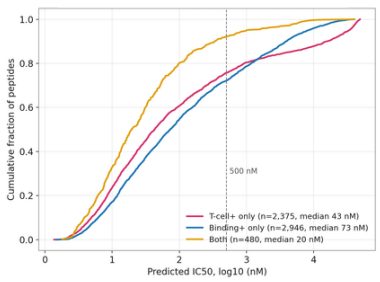


AUC = 0.88
< 500 nM

AUC = 0.89
< 50 nM

Good discrimination of binders from non-binders.

Distribution of predicted IC50 across T-cell, binding, and dual evidence groups



Median predicted IC50:

T-cell+ 43 nM

Binding+ Only 73 nM

$p \approx 1 \times 10^{-5}$ Mann-Whitney U (T-cell+ only vs binding+ only)

0.53 Probability of superiority (T-cell+ stronger)

T-cell-positive peptides were modestly enriched for strong predicted binding, but most binding-only peptides were also predicted binders.

Limitations

- Both training and test data came from IEDB, so accuracy may be overestimated.
- Assays differ across labs and formats and KD (~IC50) was treated as IC50.
- Binding data are positive-only, so the experimental range is skewed toward binders.
- Results cover HLA-A*02:01 9-mers only.
- Binding affinity does not capture processing, presentation or immunogenicity.
- Binding+ only peptides are not T-cell-negative.

Conclusions

- Predictions rank peptides well (ρ = 0.75, 95% CI 0.74–0.77).
- Typical error is about 4-fold (MAE = 0.62 log_{10} units).
- Binders and non-binders are separated well (AUC = 0.88 at 500 nM, 0.89 at 50 nM).
- T-cell+ peptides are slightly stronger predicted binders than binding+ only peptides (probability of superiority = 0.53).
- Useful for prioritising epitopes, but absolute IC_{50} is approximate.

References: *Reynisson B, Alvarez B, Paul S, Peters B, Nielsen M. NetMHCpan-4.1 and NetMHCIIpan-4.0. Nucleic Acids Res. 2020;48(W1):W449–W454. ; Vita R, et al. The Immune Epitope Database (IEDB): 2018 update. Nucleic Acids Res. 2019;47(D1):D339–D343.*